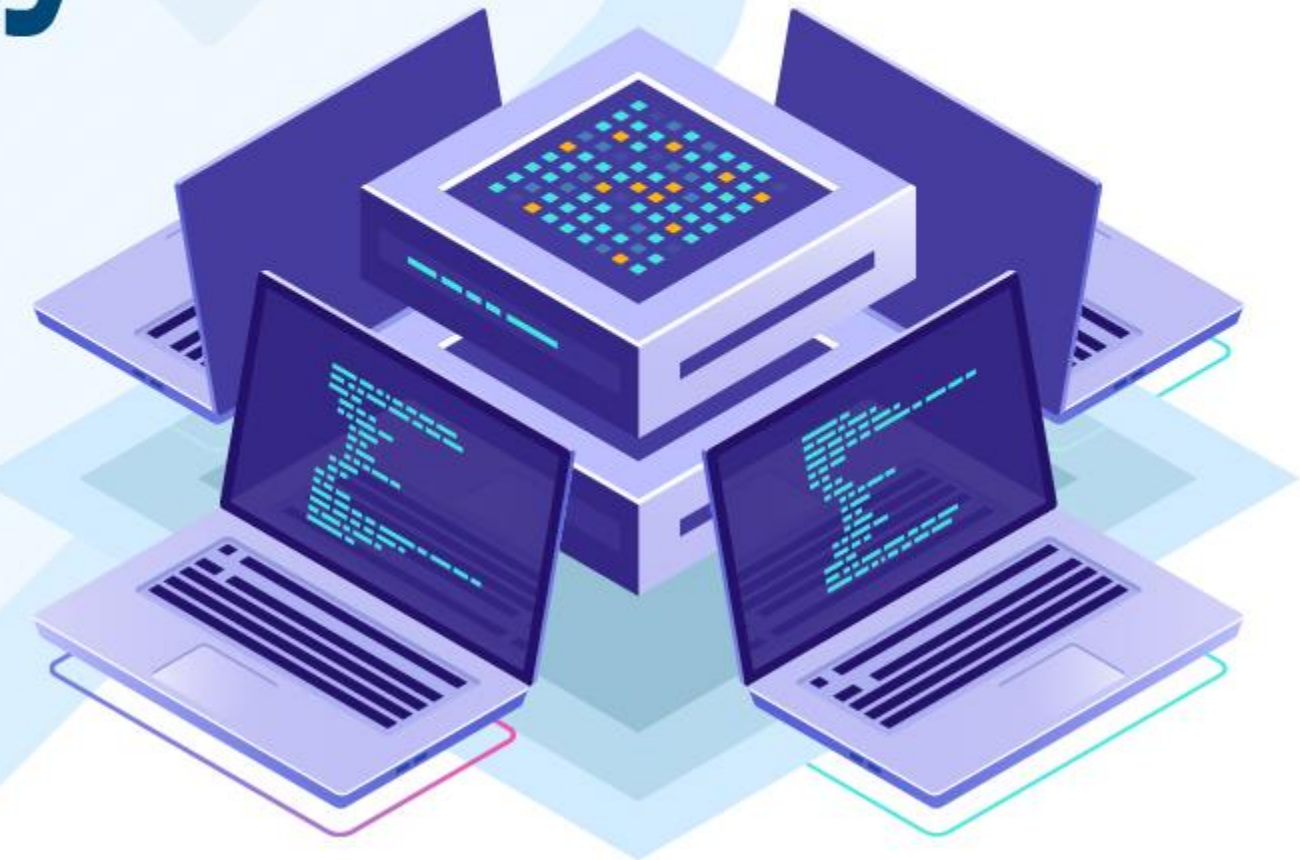
Information Technology

Year: 2026-2027

Fall Semester

- Midpoint Circle Drawing Algorithm

- **Dr. Yasmine Mahmoud**



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THE EGYPTIAN E-LEARNING UNIVERSITY

Midpoint Circle Drawing Algorithm

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The Mid-Point Circle Algorithm

- MID-POINT CIRCLE ALGORITHM

- Input radius r and circle centre (x_c, y_c), then set the coordinates for the first point on the circumference of a circle centred on the origin as:

$$(x_0, y_0) = (0, r)$$

- Calculate **the initial value of the decision parameter** as:

$$p_0 = \frac{5}{4} - r$$

- Perform the test, Starting with $k = 0$ at each position x_k, perform the following test.
- (i) If $p_k < 0$, the next point along the circle centred on $(0, 0)$ is (x_k+1, y_k) and:

$$p_{k+1} = p_k + 2x_{k+1} + 1$$

The Mid-Point Circle Algorithm (cont…)

- (ii) If Pk >0 then the next point along the circle is (x_k+1, y_k-1) and:

$$p_{k+1} = p_k + 2x_{k+1} + 1 - 2y_{k+1}$$

- Identify the symmetry points in the other seven octants
- Move (x, y) according to:

$$x = x + x_c \qquad y = y + y_c$$

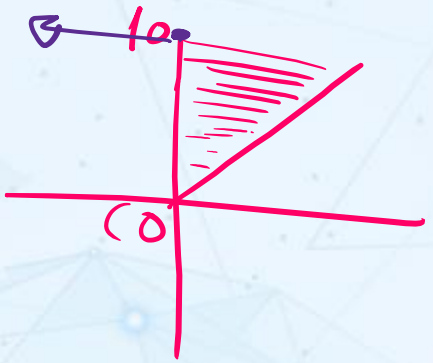
- Repeat steps 3 to 5 until $x >= y$

Mid-Point Circle Algorithm Example

- To see the mid-point circle algorithm in action lets use it to draw a circle centred at (0,0) with radius 10
- Determine the positions along the circle octant in the **first quadrant** from x=0 to x=y.
- The initial value of the decision parameter is
- $P0 = 1-r = 1-10 = -9$
- For circle centred on the coordinate origin, the initial point is (X0,Y0)=(0,10)

- Draw a circle $r = 10$

(0, 10)
(x, y)



① $P_0 = 1 - r$

$P_0 = 1 - 10 = -9$

$P_0 = -9$

iteration

K	P_k	x_{k+1}	y_{k+1}	$2x_{k+1}$	$2y_{k+1}$
0	-9	$x_k = 0$ $\therefore x_{k+1} = 1$ 1	$P_k < 0$ $\therefore y_{k+1} = y_k$ $= 10$	$2 * 1$ $= 2$	$2 * 10$ $= 20$

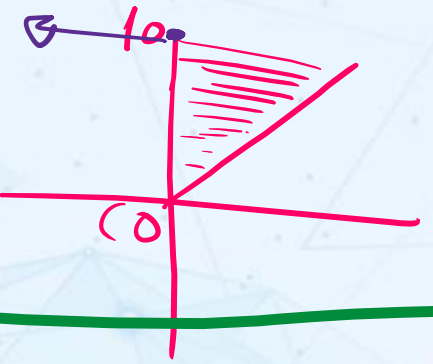
$P_k < 0 \Rightarrow -9 < 0$

$\therefore P_{k+1} = P_k + 2x_{k+1} + 1 = -9 + 2 + 1 = -6$

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- Draw a circle $r = 10$

$(0, 10)$
(x, y)



iteration

K	P_k	x_{k+1}	y_{k+1}	$2x_{k+1}$	$2y_{k+1}$
1	$P_1 = \boxed{-6}$	$x++ = \boxed{2}$	$P_k < 0$ $-6 < 0$ $\therefore \boxed{y_{k+1} = y_k}$ $= \boxed{10}$	$2 * 2 = \boxed{4}$	$2 * 10 = \boxed{20}$

$P_k < 0 \Rightarrow -6 < 0$

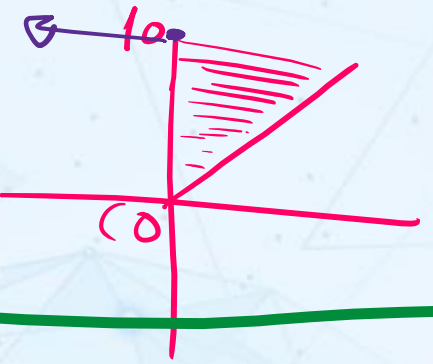
$\therefore P_{k+1} = P_k + 2x_{k+1} + 1 = -6 + 4 + 1$

$= \boxed{-1}$

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- Draw a circle $r = 10$

$(0, 10)$
(x, y)



iterate

K	P_k	x_{k+1}	y_{k+1}	$2x_{k+1}$	$2y_{k+1}$
2	-1	$x++$ = 3	$P_k < 0$ $-1 < 0$ $y_{k+1} = y_k$ = 10	$2 * 3$ = 6	$2 * 10$ = 20

$P_k < 0 \Rightarrow -1 < 0$

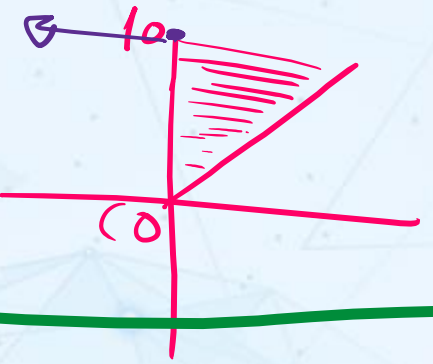
$\therefore P_{k+1} = P_k + 2x_{k+1} + 1$

$= -1 + 6 + 1 =$ 6

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- Draw a circle $r = 10$

$(0, 10)$
(x, y)



iteration

K	P_k	x_{k+1}	y_{k+1}	$2x_{k+1}$	$2y_{k+1}$
3	6	$x++$ $= \boxed{4}$	$P_k \geq 0$ $y_{k+1} = y_k - 1$ $= 10 - 1$ $= \boxed{9}$	$2 * 4$ $= \boxed{8}$	$2 * 9$ $= \boxed{18}$

$P_k \geq 0 \Rightarrow 6 \geq 0$

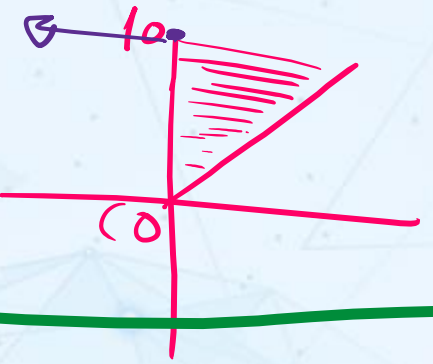
$\therefore P_{k+1} = P_k + 2x_{k+1} + 1 + 2y_{k+1}$

$= 6 + 8 + 1 - 18 = \boxed{-3}$

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- Draw a circle $r = 10$

$(0, 10)$
(x, y)



iterate

K	P_k	x_{k+1}	y_{k+1}	$2x_{k+1}$	$2y_{k+1}$
4	-3	$x++$ $= \boxed{5}$	$P_k < 0$ $y_{k+1} = y_k$ $= \boxed{9}$	2×5 $= \boxed{10}$	$2 * 9$ $= \boxed{18}$

$P_k < 0 \Rightarrow -3 < 0$

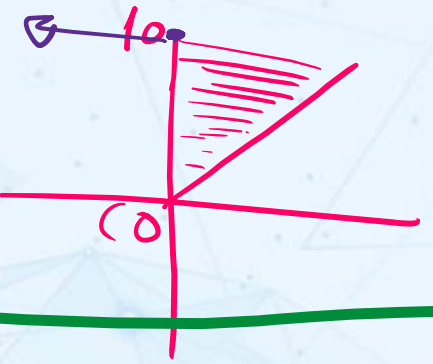
$\therefore P_{k+1} = P_k + 2x_{k+1} + 1$

$= -3 + 10 + 1 = \boxed{8}$

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- Draw a circle $r = 10$

$(0, 10)$
(x, y)



iteration

K	P_k	x_{k+1}	y_{k+1}	$2x_{k+1}$	$2y_{k+1}$
5	8	$x++$ $= \boxed{6}$	$P_k \geq 0$ $y_{k+1} = y_k - 1$ $= 9 - 1$ $= \boxed{8}$	$2 * 6$ $= \boxed{12}$	$2 * 8$ $= \boxed{16}$

$P_k \geq 0 \Rightarrow 8 \geq 0$

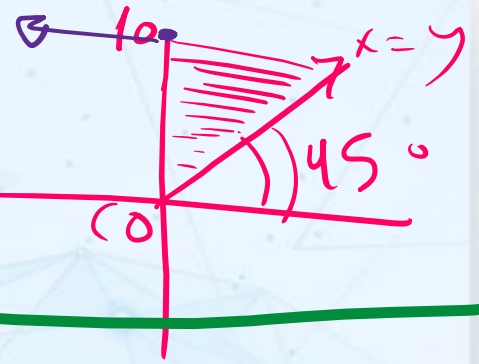
$\therefore P_{k+1} = P_k + 2x_{k+1} + 1 - 2y_{k+1}$

$= 8 + 12 + 1 - 16 = \boxed{5}$

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- Draw a circle $r = 10$

$(0, 10)$
(x, y)



iterate

K	P_k	x_{k+1}	y_{k+1}	$2x_{k+1}$	$2y_{k+1}$
6	5	$x++$ $= \boxed{7}$	$P_k \geq 0$ $y_{k+1} = y_k - 1$ $= 9 - 1$ $= \boxed{7}$	2×7 $= \boxed{14}$	2×7 $= \boxed{14}$

$\because y = x \quad (7 = 7)$

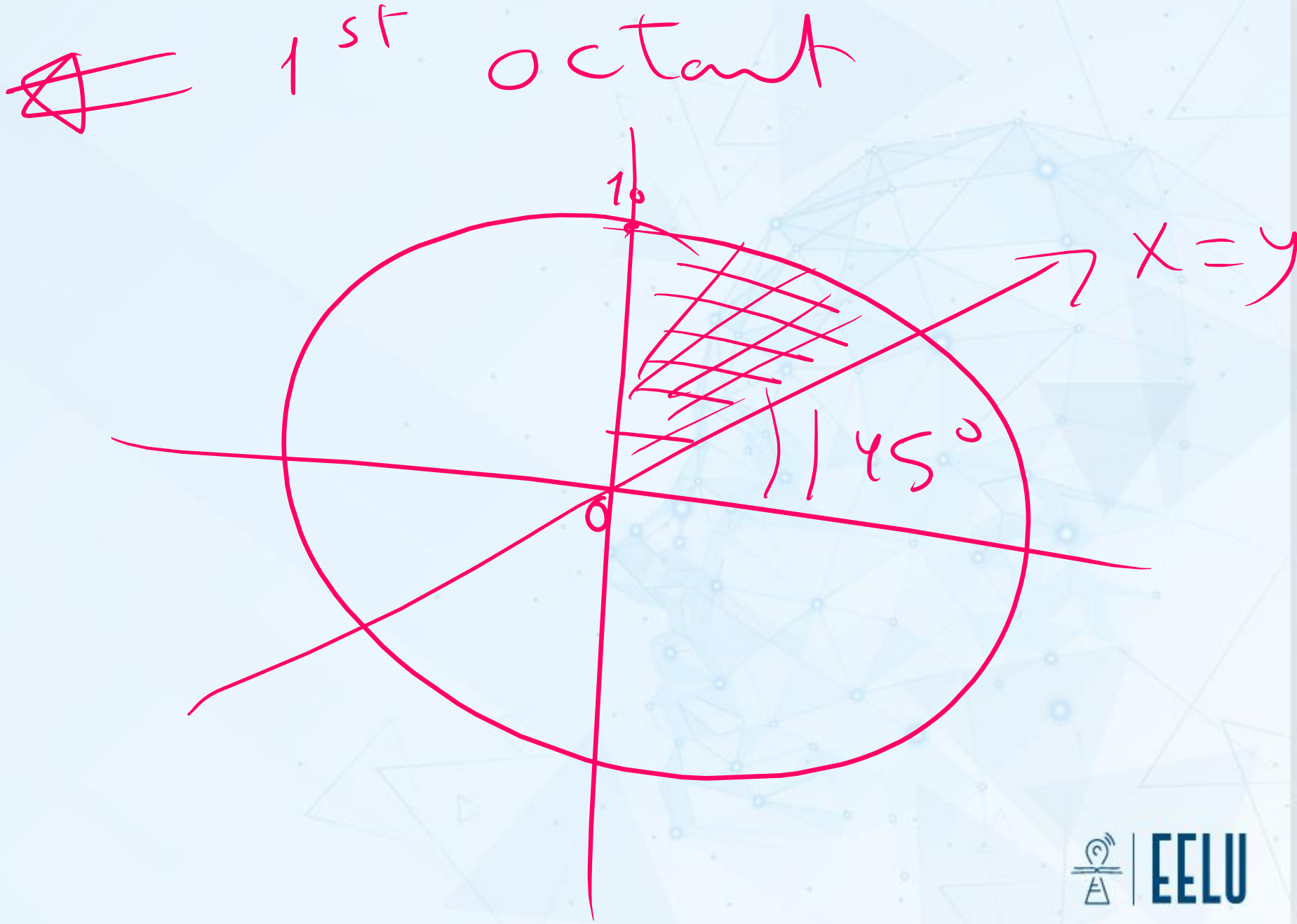
$\boxed{1^{st}\ 1/8} \Longrightarrow$ Done "stop"



Page

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X	y
0	10
1	10
2	10
3	10
4	9
5	8
6	8
7	7



1st

1/4

X	Y
0	10
1	10
2	10
3	10
4	9
5	8
6	8
x	**Y**
10	0
10	1
10	2
10	3
9	4
9	5
8	6

2nd

X (-)	Y (+)
0	10
-1	10
-2	10
-3	10
-4	9
-5	8
-6	8
x	**Y**
-10	0
-10	1
-10	2
-10	3
-9	4
-9	5
-8	6

3rd

X (-)	Y (-)
0	-10
-1	-10
-2	-10
-3	-10
-4	-9
-5	-8
-6	-8
X	**Y**
-10	0
-10	-1
-10	-2
-10	-3
-9	-4
-9	-5
-8	-6

uTh

X (+)	Y (-)
0	-10
1	-10
2	-10
3	-10
4	-9
5	-8
6	-8
x	**Y**
10	0
10	-1
10	-2
10	-3
9	-4
9	-5
8	-6

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Mid-Point Circle Algorithm Example

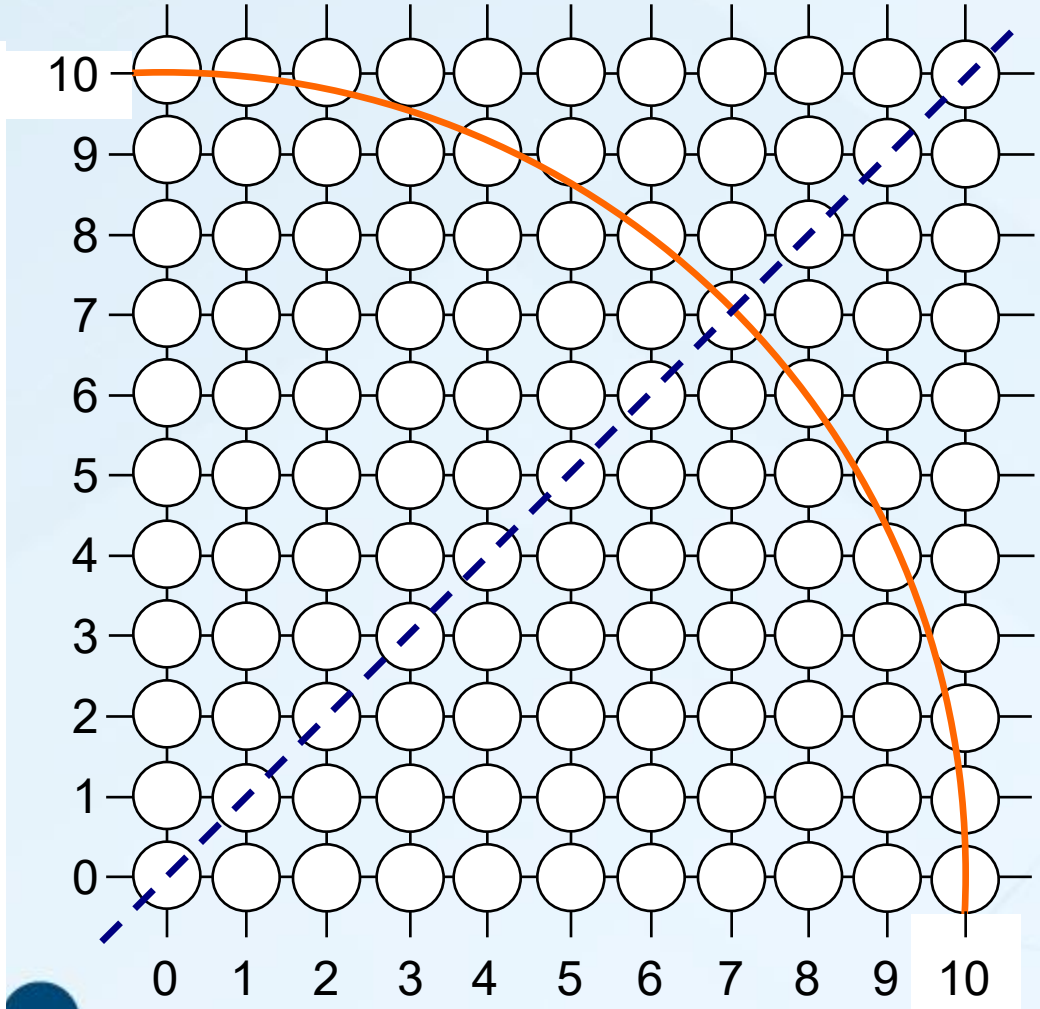
- K=0 and P0 = -9 (1,10)
- $p_{k+1} = p_k + 2x_{k+1} + 1$ (pk<0)
- K=1, P1=P0+2Xk+1 => -9+2(1)+1 =-9+3=-6 (2,10)
- K=2, P2= P1 +2(2)+1 = -6+4+1 = -1 (3,10)
- K=3 P3=P2+2(3)+1 = -1+7= 6 (4,9)

$p_{k+1} = p_k + 2x_{k+1} + 1 - 2y_{k+1}$

- K=4 (Pk>0)
- P4= P3+2(4)+1-2(9) => 6+8+1-18 = -3 (5,9)
- K=5 p5= p4+2(5)+1 => -3+10+1 = 8 (6,8)
- K=6 p6=8+2(6)+1-2(8) => 8+12+1-16 = 5 (7,7)
-

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Mid-Point Circle Algorithm Example



k	p_k	(x_{k+1},y_{k+1})	$2x_{k+1}$	$2y_{k+1}$
0	-9	(1,10)	2	20
1	-6	(2,10)	4	20
2	-1	(3,10)	6	20
3	6	(4,9)	8	18
4	-3	(5,9)	10	18
5	8	(6,8)	12	16
6	5	(7,7)	14	14

Step-05:

Step-03 is executed similarly until $X_{k+1} >= Y_{k+1}$ as follows-

P_k	P_{k+1}	(X_{k+1}, Y_{k+1})
		(0, 10)
-9	-6	(1, 10)
-6	-1	(2, 10)
-1	6	(3, 10)
6	-3	(4, 9)
-3	8	(5, 9)
8	5	(6, 8)
Algorithm Terminates **These are all points for Octant-1.**		

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Algorithm calculates all the points of octant-1 and terminates.

Now, the points of octant-2 are obtained using the mirror effect by swapping X and Y coordinates.

Octant-1 Points	Octant-2 Points
(0, 10)	(8, 6)
(1, 10)	(9, 5)
(2, 10)	(9, 4)
(3, 10)	(10, 3)
(4, 9)	(10, 2)
(5, 9)	(10, 1)
(6, 8)	(10, 0)
These are all points for Quadrant-1.	

Quadrant-1 (X,Y)	Quadrant-2 (-X,Y)	Quadrant-3 (-X,-Y)	Quadrant-4 (X,-Y)
(0, 10)	(0, 10)	(0, -10)	(0, -10)
(1, 10)	(-1, 10)	(-1, -10)	(1, -10)
(2, 10)	(-2, 10)	(-2, -10)	(2, -10)
(3, 10)	(-3, 10)	(-3, -10)	(3, -10)
(4, 9)	(-4, 9)	(-4, -9)	(4, -9)
(5, 9)	(-5, 9)	(-5, -9)	(5, -9)
(6, 8)	(-6, 8)	(-6, -8)	(6, -8)
(8, 6)	(-8, 6)	(-8, -6)	(8, -6)
(9, 5)	(-9, 5)	(-9, -5)	(9, -5)
(9, 4)	(-9, 4)	(-9, -4)	(9, -4)
(10, 3)	(-10, 3)	(-10, -3)	(10, -3)
(10, 2)	(-10, 2)	(-10, -2)	(10, -2)
(10, 1)	(-10, 1)	(-10, -1)	(10, -1)
(10, 0)	(-10, 0)	(-10, 0)	(10, 0)
These are all points of the Circle.			

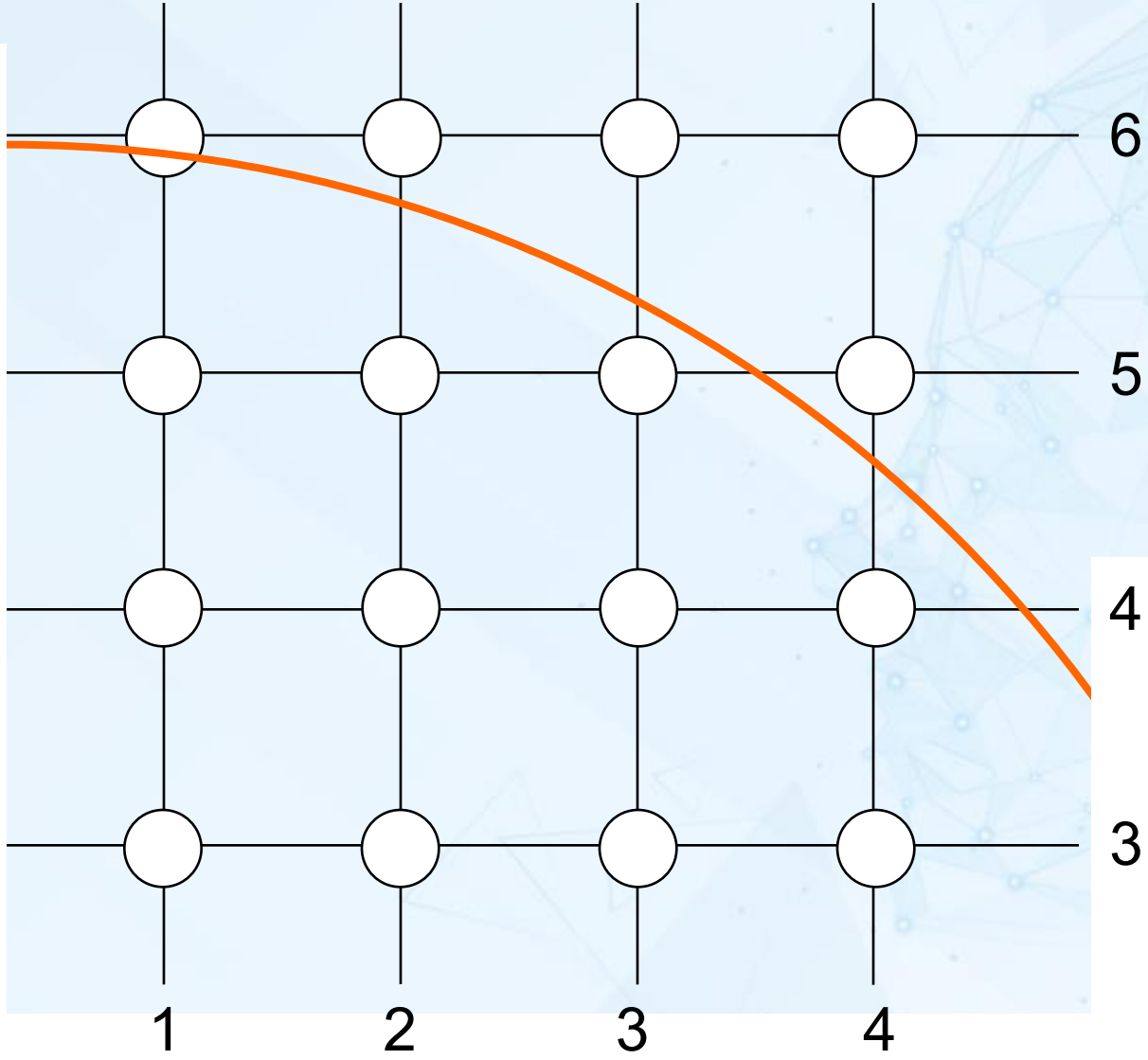
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Mid-Point Circle Algorithm Summary

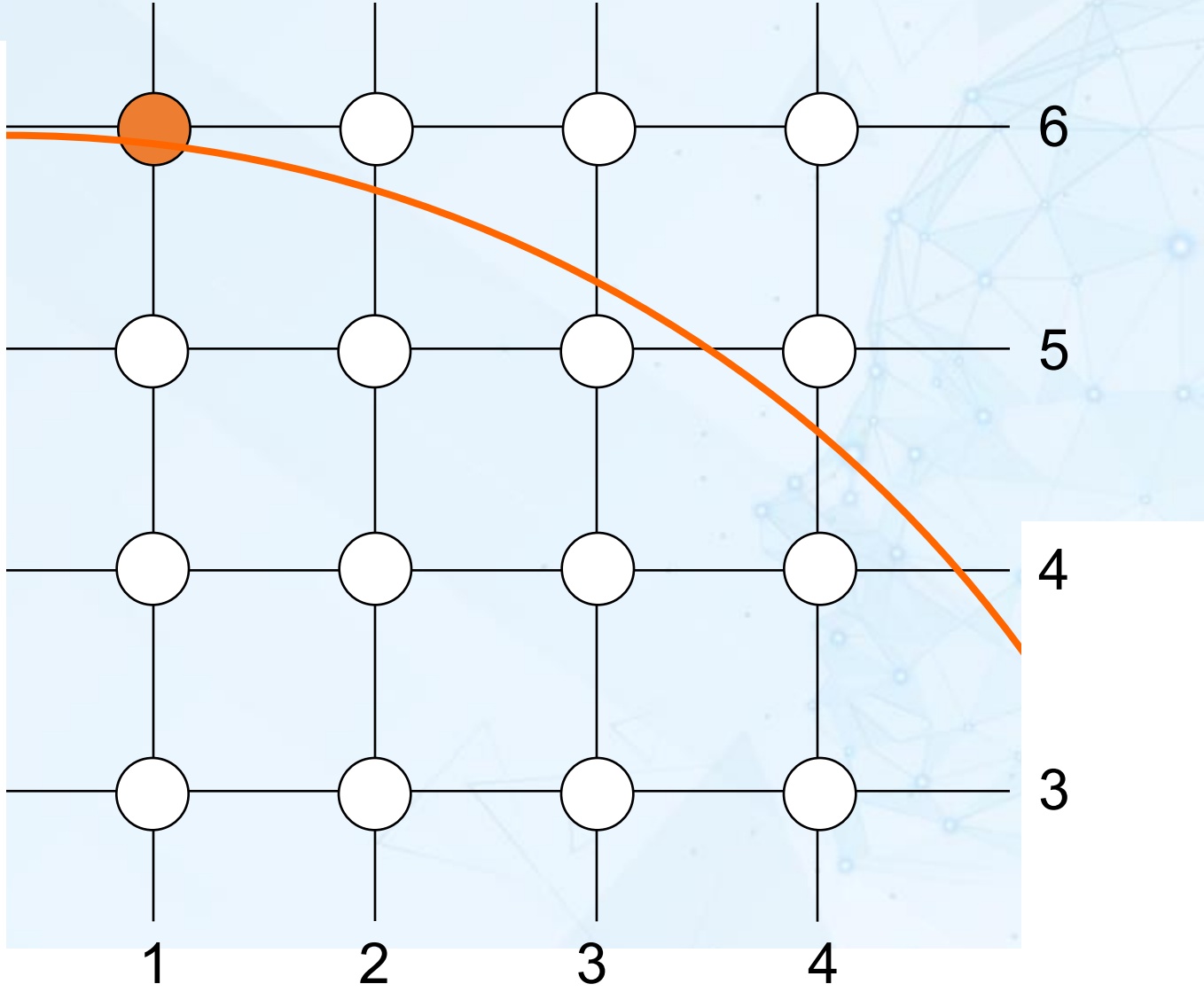
- The key insights in the **mid-point circle algorithm** are:
  - Eight-way symmetry can hugely reduce the work in drawing a circle
  - Moving in unit steps along the x axis at each point along the circle's edge we need to choose between two possible y coordinates

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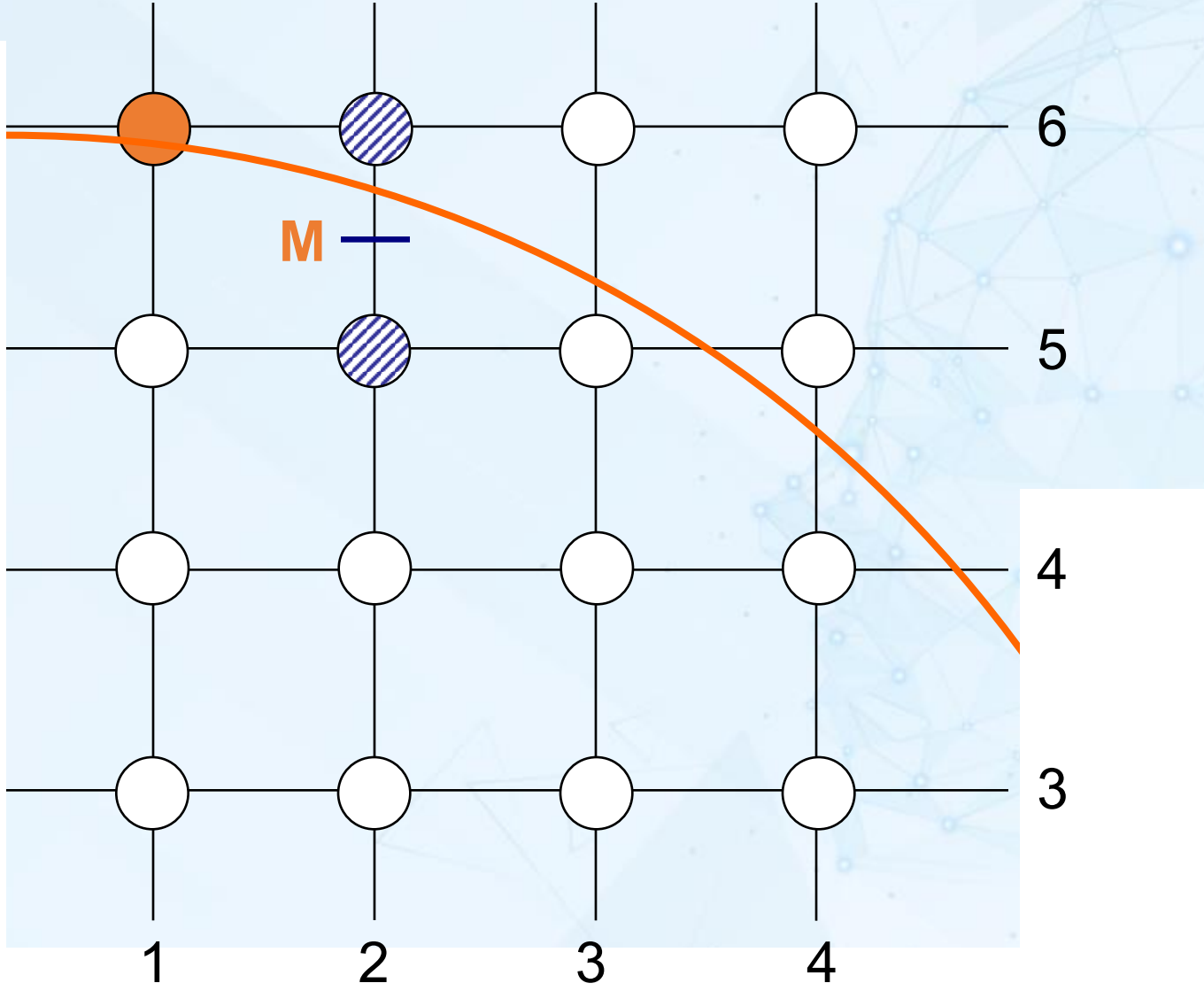
Mid-Point Circle Algorithm



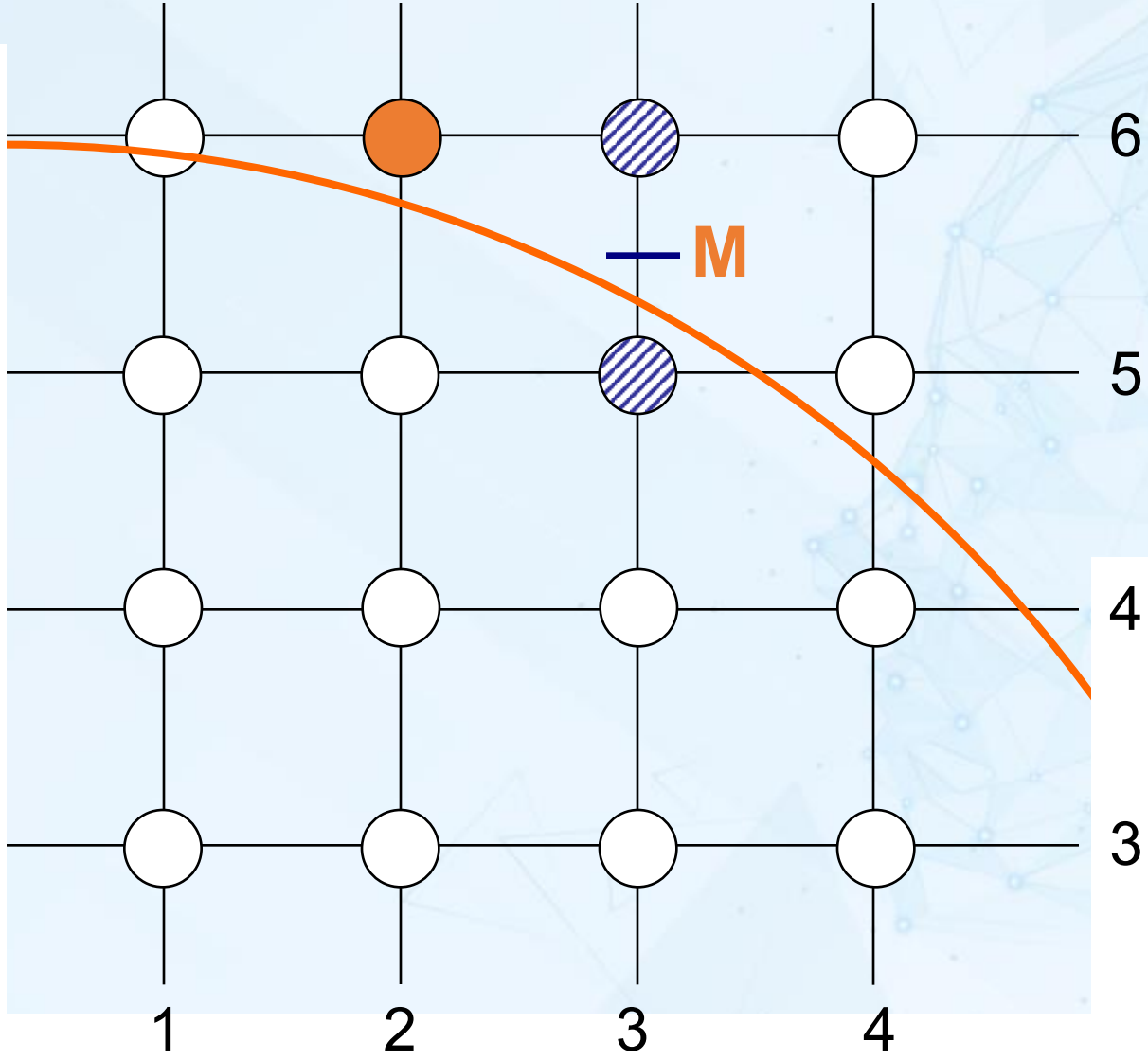
Mid-Point Circle Algorithm



Mid-Point Circle Algorithm



Mid-Point Circle Algorithm



Mid-Point Circle Algorithm

